

Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Technology, Pune-37



Department of Computer Engineering

Class and Division	CS-A
Batch Number	1
GR-No	1251070524
Roll Number	2070120
Student Name	Samiksha Sanjay Adake
Assignment Number	1
Assignment Title	Design and implement a Hospital Management System database using SQL. The system should store information about patients, doctors, and their appointments.
CO Mapped	

Theory:

Concepts used:

Database: A collection of related data stored in an organized manner.

Table: Stores information in rows and columns.

Primary Key: Uniquely identifies each record in a table.

Auto Increment: Automatically generates unique ID values.

Not Null: Ensures that a column cannot contain NULL values.

Unique: Ensures duplicate values are not allowed.

Foreign Key: Creates a relationship between two tables.

Insert: Adds new records into a table.

Update: Modifies existing records.

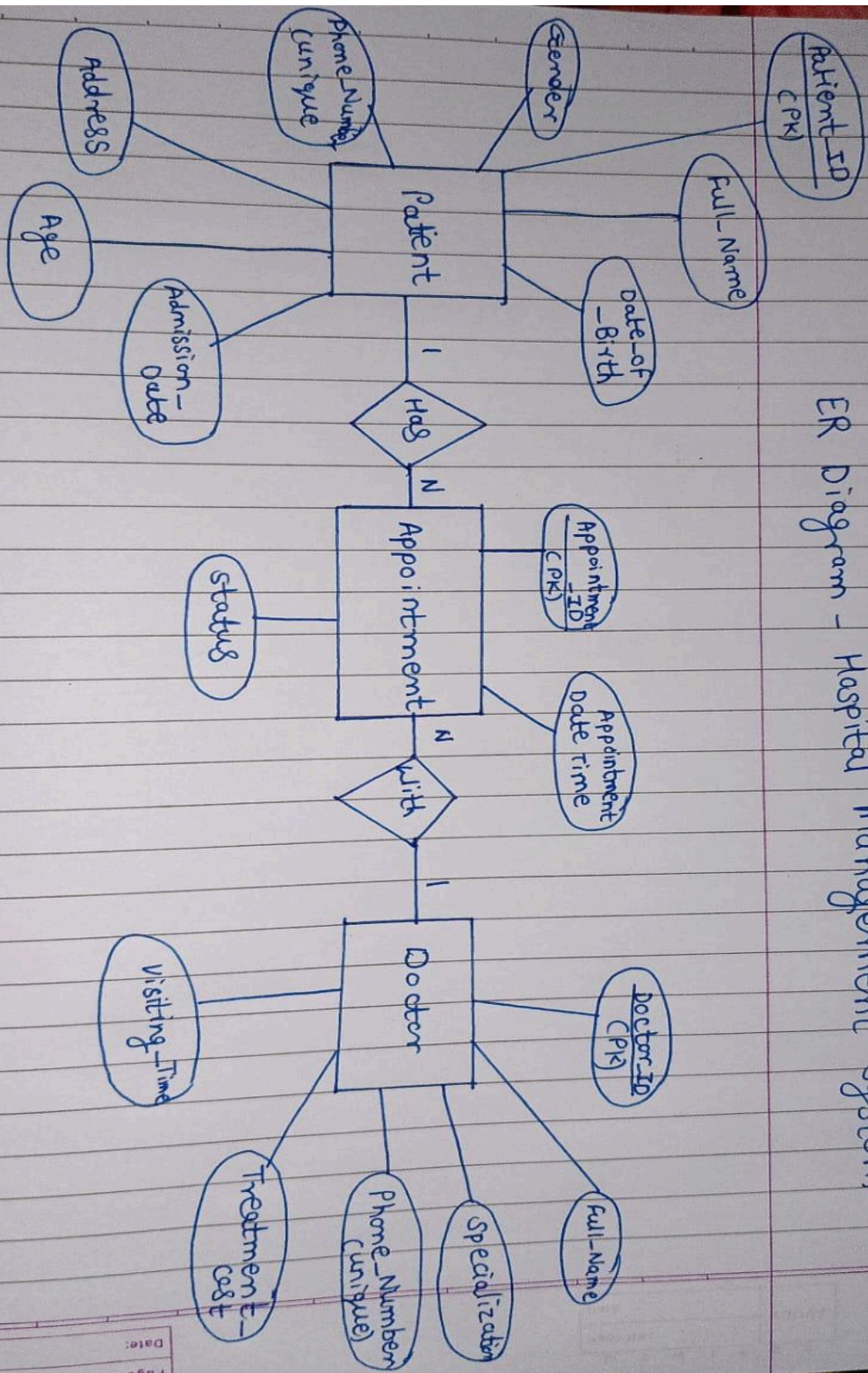
Delete: Removes records from a table.

Alter: Changes the table structure by adding, modifying, or dropping columns.

Order By: Sorts records in ascending or descending order.

Aggregate Functions: Functions such as **MAX()** and **COUNT()** are used to summarize data.

ER Diagram - Hospital Management System



Input:

Relations:

Doctor table, Patient Table, Appointment Table

SQL Queries Used

- CREATE DATABASE
- USE DATABASE
- CREATE TABLE
- INSERT
- SELECT
- UPDATE
- DELETE
- ALTER TABLE
- MODIFY
- DROP COLUMN
- ORDER BY
- WHERE
- MAX()
- COUNT()

Output:

Enter password: *****

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 8

Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE hospital_assignment;

Query OK, 1 row affected (0.07 sec)

mysql> USE hospital_assignment;

Database changed

mysql> CREATE TABLE patient (

-> Patient_ID INT AUTO_INCREMENT PRIMARY KEY,

-> Full_Name VARCHAR(100) NOT NULL,

-> Date_of_Birth DATE NOT NULL,

-> Gender VARCHAR(10) NOT NULL,

-> Phone_Number VARCHAR(15) NOT NULL UNIQUE,

-> Address VARCHAR(100) NOT NULL

->);

Query OK, 0 rows affected (0.12 sec)

mysql> DESC patient;

Field	Type	Null	Key	Default	Extra
Patient_ID	int	NO	PRI	NULL	auto_increment

Full_Name	varchar(100)	NO		NULL	
Date_of_Birth	date	NO		NULL	
Gender	varchar(10)	NO		NULL	
Phone_Number	varchar(15)	NO	UNI	NULL	
Address	varchar(100)	NO		NULL	

6 rows in set (0.06 sec)

```
mysql> CREATE TABLE doctor (
->   Doctor_ID INT AUTO_INCREMENT PRIMARY KEY,
->   Full_Name VARCHAR(100) NOT NULL,
->   Specialization VARCHAR(50) NOT NULL,
->   Phone_Number VARCHAR(15) NOT NULL UNIQUE,
->   Treatment_Cost DECIMAL(10,2) NOT NULL,
->   Department VARCHAR(50) NOT NULL
-> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> DESC doctor;
```

Field	Type	Null	Key	Default	Extra
Doctor_ID	int	NO	PRI	NULL	auto_increment
Full_Name	varchar(100)	NO		NULL	
Specialization	varchar(50)	NO		NULL	
Phone_Number	varchar(15)	NO	UNI	NULL	
Treatment_Cost	decimal(10,2)	NO		NULL	
Department	varchar(50)	NO		NULL	

6 rows in set (0.00 sec)

```
mysql> CREATE TABLE appointment (
->   Appointment_ID INT AUTO_INCREMENT PRIMARY KEY,
->   Patient_ID INT NOT NULL,
->   Doctor_ID INT NOT NULL,
->   Appointment_DateTime DATETIME NOT NULL,
->   Status VARCHAR(20) NOT NULL,
->   FOREIGN KEY (Patient_ID) REFERENCES patient(Patient_ID),
->   FOREIGN KEY (Doctor_ID) REFERENCES doctor(Doctor_ID)
-> );
```

Query OK, 0 rows affected (0.05 sec)

mysql> DESC appointment;

Field	Type	Null	Key	Default	Extra
Appointment_ID	int	NO	PRI	NULL	auto_increment
Patient_ID	int	NO	MUL	NULL	
Doctor_ID	int	NO	MUL	NULL	
Appointment_DateTime	datetime	NO		NULL	
Status	varchar(20)	NO		NULL	

5 rows in set (0.00 sec)

mysql> INSERT INTO patient (Full_Name, Date_of_Birth, Gender, Phone_Number, Address)

-> VALUES

-> ('Rahul Sharma','2001-05-15','Male','9876543210','Pune'),
-> ('Priya Patil','1994-08-22','Female','9876501234','Mumbai'),
-> ('Amit Kumar','1979-02-10','Male','9876512345','Kolhapur'),
-> ('Sneha Joshi','1996-11-05','Female','9876523456','Satara'),
-> ('Rohan Desai','1987-07-18','Male','9876534567','Sangli'),
-> ('Anjali Kulkarni','1998-03-25','Female','9876545678','Nashik'),
-> ('Vikram Singh','1974-09-14','Male','9876556789','Nagpur'),
-> ('Neha Verma','1991-12-30','Female','9876567890','Solapur'),
-> ('Karan Mehta','1984-04-08','Male','9876578901','Aurangabad'),
-> ('Pooja More','1993-10-12','Female','9876589012','Ahmednagar');

Query OK, 10 rows affected (0.03 sec)

Records: 10 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM patient;

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address
1	Rahul Sharma	2001-05-15	Male	9876543210	Pune
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli

6	Anjali Kulkarni	1998-03-25	Female	9876545678	Nashik
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur
8	Neha Verma	1991-12-30	Female	9876567890	Solapur
9	Karan Mehta	1984-04-08	Male	9876578901	Aurangabad
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar

10 rows in set (0.01 sec)

```
mysql> INSERT INTO doctor (Full_Name, Specialization, Phone_Number,
Treatment_Cost, Department)
-> VALUES
-> ('Dr. Sharma','Cardiology','9876600001',5000.00,'Heart'),
-> ('Dr. Mehta','Orthopedic','9876600002',3500.00,'Bone'),
-> ('Dr. Patil','Neurology','9876600003',6500.00,'Brain'),
-> ('Dr. Joshi','Dermatology','9876600004',2500.00,'Skin'),
-> ('Dr. Kulkarni','Cardiology','9876600005',5500.00,'Heart'),
-> ('Dr. Deshmukh','Pediatrics','9876600006',3000.00,'Children'),
-> ('Dr. Shah','ENT','9876600007',2800.00,'ENT'),
-> ('Dr. Reddy','General Medicine','9876600008',2000.00,'Medicine');
Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM doctor;
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Department
1	Dr. Sharma	Cardiology	9876600001	5000.00	Heart
2	Dr. Mehta	Orthopedic	9876600002	3500.00	Bone
3	Dr. Patil	Neurology	9876600003	6500.00	Brain
4	Dr. Joshi	Dermatology	9876600004	2500.00	Skin
5	Dr. Kulkarni	Cardiology	9876600005	5500.00	Heart
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00	Children
7	Dr. Shah	ENT	9876600007	2800.00	ENT
8	Dr. Reddy	General Medicine	9876600008	2000.00	Medicine

8 rows in set (0.01 sec)

```
mysql> INSERT INTO appointment (Patient_ID, Doctor_ID,
Appointment_DateTime, Status)
-> VALUES
-> (1,1,'2026-08-02 09:00:00','Scheduled'),
-> (2,2,'2026-08-02 10:00:00','Completed'),
-> (3,3,'2026-08-03 11:00:00','Scheduled'),
-> (4,4,'2026-08-03 12:00:00','Cancelled'),
-> (5,5,'2026-08-04 09:30:00','Scheduled'),
-> (6,6,'2026-08-04 10:30:00','Completed'),
-> (7,7,'2026-08-05 11:30:00','Scheduled'),
-> (8,8,'2026-08-05 12:30:00','Cancelled'),
-> (9,1,'2026-08-06 09:15:00','Scheduled'),
-> (10,2,'2026-08-06 10:15:00','Scheduled');
```

Query OK, 10 rows affected (0.02 sec)

Records: 10 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM appointment;
```

Appointment_ID	Patient_ID	Doctor_ID	Appointment_DateTime	Status
1	1	1	2026-08-02 09:00:00	Scheduled
2	2	2	2026-08-02 10:00:00	Completed
3	3	3	2026-08-03 11:00:00	Scheduled
4	4	4	2026-08-03 12:00:00	Cancelled
5	5	5	2026-08-04 09:30:00	Scheduled
6	6	6	2026-08-04 10:30:00	Completed
7	7	7	2026-08-05 11:30:00	Scheduled
8	8	8	2026-08-05 12:30:00	Cancelled
9	9	1	2026-08-06 09:15:00	Scheduled
10	10	2	2026-08-06 10:15:00	Scheduled

10 rows in set (0.00 sec)

```
mysql> DELETE FROM appointment
```

```
-> WHERE Patient_ID = 9;
```

Query OK, 1 row affected (0.01 sec)

```
mysql> DELETE FROM patient
```

```
-> WHERE Patient_ID = 9;
```

Query OK, 1 row affected (0.01 sec)

mysql> SELECT * FROM patient;

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address
1	Rahul Sharma	2001-05-15	Male	9876543210	Pune
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli
6	Anjali Kulkarni	1998-03-25	Female	9876545678	Nashik
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur
8	Neha Verma	1991-12-30	Female	9876567890	Solapur
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar

9 rows in set (0.00 sec)

mysql> select * from patient order by age desc;

ERROR 1054 (42S22): Unknown column 'age' in 'order clause'

mysql> ALTER TABLE patient

-> ADD Age INT;

Query OK, 0 rows affected (0.03 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql>

mysql> UPDATE patient

-> SET Age = TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE());

Query OK, 9 rows affected (0.01 sec)

Rows matched: 9 Changed: 9 Warnings: 0

mysql> SELECT *

-> FROM patient

-> ORDER BY Age DESC;

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	

```

51 |
|      3 | Amit Kumar      | 1979-02-10 | Male | 9876512345 | Kolhapur |
47 |
|      5 | Rohan Desai   | 1987-07-18 | Male | 9876534567 | Sangli   | 39 |
|      8 | Neha Verma    | 1991-12-30 | Female | 9876567890 | Solapur  |
34 |
|     10 | Pooja More    | 1993-10-12 | Female | 9876589012 | Ahmednagar |
32 |
|      2 | Priya Patil   | 1994-08-22 | Female | 9876501234 | Mumbai   | 31 |
|
|      4 | Sneha Joshi   | 1996-11-05 | Female | 9876523456 | Satara    | 29 |
|      6 | Anjali Kulkarni | 1998-03-25 | Female | 9876545678 | Nashik    |
28 |
|      1 | Rahul Sharma  | 2001-05-15 | Male | 9876543210 | Pune      | 25 |
|
+-----+-----+-----+-----+-----+-----+
9 rows in set (0.00 sec)

```

```

mysql> SELECT *
-> FROM doctor
-> ORDER BY Treatment_Cost DESC;
+-----+-----+-----+-----+-----+-----+
| Doctor_ID | Full_Name      | Specialization      | Phone_Number      |
Treatment_Cost | Department |
+-----+-----+-----+-----+-----+-----+
|      3 | Dr. Patil      | Neurology           | 9876600003        | 6500.00 | Brain      |
|      5 | Dr. Kulkarni   | Cardiology           | 9876600005        | 5500.00 | Heart      |
|      1 | Dr. Sharma     | Cardiology           | 9876600001        | 5000.00 | Heart      |
|      2 | Dr. Mehta      | Orthopedic           | 9876600002        | 3500.00 | Bone       |
|      6 | Dr. Deshmukh   | Pediatrics           | 9876600006        | 3000.00 | Children   |
|
|      7 | Dr. Shah       | ENT                  | 9876600007        | 2800.00 | ENT        |
|      4 | Dr. Joshi      | Dermatology          | 9876600004        | 2500.00 | Skin       |
|      8 | Dr. Reddy      | General Medicine     | 9876600008        | 2000.00 |
Medicine |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

```

```

mysql> update patient set Phone_Number="9999999999" where
Patient_ID=1;

```

Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from patient;

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39
6	Anjali Kulkarni	1998-03-25	Female	9876545678	Nashik	28
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32

9 rows in set (0.00 sec)

mysql> ALTER TABLE doctor

-> DROP COLUMN Doctor_ID;

ERROR 1829 (HY000): Cannot drop column 'Doctor_ID': needed in a foreign key constraint 'appointment_ibfk_2' of table 'appointment'

mysql> ALTER TABLE doctor

-> DROP COLUMN Department;

Query OK, 0 rows affected (0.02 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from doctor;

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost
-----------	-----------	----------------	--------------	----------------

1	Dr. Sharma	Cardiology	9876600001	5000.00
2	Dr. Mehta	Orthopedic	9876600002	3500.00
3	Dr. Patil	Neurology	9876600003	6500.00
4	Dr. Joshi	Dermatology	9876600004	2500.00
5	Dr. Kulkarni	Cardiology	9876600005	5500.00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00
7	Dr. Shah	ENT	9876600007	2800.00
8	Dr. Reddy	General Medicine	9876600008	2000.00

8 rows in set (0.00 sec)

```
mysql> update patient set Address='Nagpur',Phone_Number='8888888888'
where Patient_ID=6;
```

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> select * from patient
```

-> ;

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39
6	Anjali Kulkarni	1998-03-25	Female	8888888888	Nagpur	28
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32

9 rows in set (0.00 sec)

mysql> UPDATE doctor

-> SET Treatment_Cost = Treatment_Cost + 100

-> WHERE Specialization='ENT';

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from doctor;

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost
1	Dr. Sharma	Cardiology	9876600001	5000.00
2	Dr. Mehta	Orthopedic	9876600002	3500.00
3	Dr. Patil	Neurology	9876600003	6500.00
4	Dr. Joshi	Dermatology	9876600004	2500.00
5	Dr. Kulkarni	Cardiology	9876600005	5500.00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00
7	Dr. Shah	ENT	9876600007	2900.00
8	Dr. Reddy	General Medicine	9876600008	2000.00

8 rows in set (0.00 sec)

mysql> ALTER TABLE patient

-> ADD Admission_Date DATE;

Query OK, 0 rows affected (0.02 sec)

Records: 0 Duplicates: 0 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-01' WHERE Patient_ID=1;

Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-03' WHERE Patient_ID=2;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-05' WHERE

Patient_ID=3;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-08' WHERE
Patient_ID=4;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-10' WHERE
Patient_ID=5;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-12' WHERE
Patient_ID=6;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-15' WHERE
Patient_ID=7;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-18' WHERE
Patient_ID=8;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-20' WHERE
Patient_ID=9;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-22' WHERE
Patient_ID=10;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM patient;

```

+-----+-----+-----+-----+-----+-----+-----+
| Patient_ID | Full_Name      | Date_of_Birth | Gender | Phone_Number | Address | Age | Admission_Date |
+-----+-----+-----+-----+-----+-----+-----+
|      1 | Rahul Sharma   | 2001-05-15    | Male   | 9999999999    | Pune    | 25 | 2026-07-01    |
|      2 | Priya Patil    | 1994-08-22    | Female | 9876501234    | Mumbai  | 31 | 2026-07-03    |
|      3 | Amit Kumar     | 1979-02-10    | Male   | 9876512345    | Kolhapur | 47 | 2026-07-05    |
|      4 | Sneha Joshi    | 1996-11-05    | Female | 9876523456    | Satara   | 29 | 2026-07-08    |
|      5 | Rohan Desai    | 1987-07-18    | Male   | 9876534567    | Sangli   | 39 | 2026-07-10    |
|      6 | Anjali Kulkarni | 1998-03-25    | Female | 8888888888    | Nagpur   | 28 | 2026-07-12    |
|      7 | Vikram Singh    | 1974-09-14    | Male   | 9876556789    | Nagpur   | 51 | 2026-07-15    |
|      8 | Neha Verma     | 1991-12-30    | Female | 9876567890    | Solapur  | 34 | 2026-07-18    |
|     10 | Pooja More     | 1993-10-12    | Female | 9876589012    | Ahmednagar | 32 | 2026-07-22    |
+-----+-----+-----+-----+-----+-----+-----+

```

9 rows in set (0.00 sec)

```
mysql> alter table doctor add Visiting_Time TIME;
```

Query OK, 0 rows affected (0.02 sec)

Records: 0 Duplicates: 0 Warnings: 0

```
mysql> UPDATE doctor SET Visiting_Time='09:00:00' WHERE Doctor_ID=1;
```

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> UPDATE doctor SET Visiting_Time='09:30:00' WHERE Doctor_ID=2;
```

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='10:00:00' WHERE Doctor_ID=3;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='10:30:00' WHERE Doctor_ID=4;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='11:00:00' WHERE Doctor_ID=5;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='11:30:00' WHERE Doctor_ID=6;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='12:00:00' WHERE Doctor_ID=7;

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='12:30:00' WHERE Doctor_ID=8;

Query OK, 1 row affected (0.01 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from doctor;

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Visiting_Time
1	Dr. Sharma	Cardiology	9876600001	5000.00	09:00:00
2	Dr. Mehta	Orthopedic	9876600002	3500.00	09:30:00

3	Dr. Patil	Neurology	9876600003	6500.00	10:00:00
4	Dr. Joshi	Dermatology	9876600004	2500.00	10:30:00
5	Dr. Kulkarni	Cardiology	9876600005	5500.00	11:00:00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00	11:30:00
7	Dr. Shah	ENT	9876600007	2900.00	12:00:00
8	Dr. Reddy	General Medicine	9876600008	2000.00	12:30:00

8 rows in set (0.00 sec)

```
mysql> ALTER TABLE patient
-> MODIFY Phone_Number VARCHAR(20);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql> select * from patient;
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25	2026-07-01
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31	2026-07-03
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47	2026-07-05
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29	2026-07-08
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39	2026-07-10
6	Anjali Kulkarni	1998-03-25	Female	8888888888	Nagpur	28	2026-07-12
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51	2026-07-15

```

|      8 | Neha Verma      | 1991-12-30   | Female | 9876567890 | Solapur   |
34 | 2026-07-18      |
|      10 | Pooja More      | 1993-10-12   | Female | 9876589012 | Ahmednagar |
32 | 2026-07-22      |
+-----+-----+-----+-----+-----+-----+
-----+
9 rows in set (0.00 sec)

```

mysql>

mysql> select from patient where age>30;

ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'from patient where age>30' at line 1

mysql> select * from patient where age>30;

```

+-----+-----+-----+-----+-----+-----+
-----+
| Patient_ID | Full_Name      | Date_of_Birth | Gender | Phone_Number | Address    | Age | Admission_Date |
+-----+-----+-----+-----+-----+-----+
-----+
|      2 | Priya Patil   | 1994-08-22   | Female | 9876501234   | Mumbai    | 31 | 2026-07-03     |
|      3 | Amit Kumar    | 1979-02-10   | Male   | 9876512345   | Kolhapur  | 47 | 2026-07-05     |
|      5 | Rohan Desai   | 1987-07-18   | Male   | 9876534567   | Sangli    | 39 | 2026-07-10     |
|      7 | Vikram Singh  | 1974-09-14   | Male   | 9876556789   | Nagpur    | 51 | 2026-07-15     |
|      8 | Neha Verma    | 1991-12-30   | Female | 9876567890   | Solapur   | 34 | 2026-07-18     |
|     10 | Pooja More    | 1993-10-12   | Female | 9876589012   | Ahmednagar | 32 | 2026-07-22     |
+-----+-----+-----+-----+-----+-----+
-----+
6 rows in set (0.00 sec)

```

mysql> select Full_Name from doctor where Treatment_Cost >2000;

```

+-----+
| Full_Name |
+-----+

```

Dr. Sharma
Dr. Mehta
Dr. Patil
Dr. Joshi
Dr. Kulkarni
Dr. Deshmukh
Dr. Shah

+-----+

7 rows in set (0.01 sec)

```
mysql> SELECT COUNT(*) AS Total_Patients
-> FROM patient;
```

+-----+

Total_Patients

+-----+

9

+-----+

1 row in set (0.01 sec)

```
mysql> SELECT MAX(Treatment_Cost) AS Maximum_Treatment_Cost
-> FROM doctor;
```

+-----+

Maximum_Treatment_Cost

+-----+

6500.00

+-----+

1 row in set (0.01 sec)

```
mysql> select * from patient order by age desc;
```

+-----+-----+-----+-----+-----+-----+-----+-----+
-----+

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
------------	-----------	---------------	--------	--------------	---------	-----	----------------

+-----+-----+-----+-----+-----+-----+-----+-----+
-----+

7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51	2026-07-15
---	--------------	------------	------	------------	--------	----	------------

3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47	2026-07-05
---	------------	------------	------	------------	----------	----	------------

5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39	
---	-------------	------------	------	------------	--------	----	--

```

2026-07-10 |
|      8 | Neha Verma      | 1991-12-30 | Female | 9876567890 | Solapur |
34 | 2026-07-18 |
|     10 | Pooja More      | 1993-10-12 | Female | 9876589012 | Ahmednagar |
32 | 2026-07-22 |
|      2 | Priya Patil     | 1994-08-22 | Female | 9876501234 | Mumbai   | 31
| 2026-07-03 |
|      4 | Sneha Joshi     | 1996-11-05 | Female | 9876523456 | Satara   | 29 |
2026-07-08 |
|      6 | Anjali Kulkarni | 1998-03-25 | Female | 8888888888 | Nagpur   |
28 | 2026-07-12 |
|      1 | Rahul Sharma    | 2001-05-15 | Male   | 9999999999 | Pune     | 25
| 2026-07-01 |

```

```

+-----+-----+-----+-----+-----+-----+-----+
-----+

```

9 rows in set (0.00 sec)

```

mysql> UPDATE appointment
-> SET Status='Completed'
-> WHERE Appointment_ID=1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

```

```

mysql>
mysql> SELECT * FROM appointment;
+-----+-----+-----+-----+-----+
| Appointment_ID | Patient_ID | Doctor_ID | Appointment_DateTime | Status |
+-----+-----+-----+-----+-----+
|      1 |      1 |      1 | 2026-08-02 09:00:00 | Completed |
|      2 |      2 |      2 | 2026-08-02 10:00:00 | Completed |
|      3 |      3 |      3 | 2026-08-03 11:00:00 | Scheduled |
|      4 |      4 |      4 | 2026-08-03 12:00:00 | Cancelled |
|      5 |      5 |      5 | 2026-08-04 09:30:00 | Scheduled |
|      6 |      6 |      6 | 2026-08-04 10:30:00 | Completed |
|      7 |      7 |      7 | 2026-08-05 11:30:00 | Scheduled |
|      8 |      8 |      8 | 2026-08-05 12:30:00 | Cancelled |
|     10 |     10 |      2 | 2026-08-06 10:15:00 | Scheduled |
+-----+-----+-----+-----+-----+

```

9 rows in set (0.00 sec)

```
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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```

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> CREATE DATABASE hospital_assignment;
Query OK, 1 row affected (0.07 sec)
```

```
mysql> USE hospital_assignment;
Database changed
```

```
mysql> CREATE TABLE patient (
  -> Patient_ID INT AUTO_INCREMENT PRIMARY KEY,
  -> Full_Name VARCHAR(100) NOT NULL,
  -> Date_of_Birth DATE NOT NULL,
  -> Gender VARCHAR(10) NOT NULL,
  -> Phone_Number VARCHAR(15) NOT NULL UNIQUE,
  -> Address VARCHAR(100) NOT NULL
  -> );
Query OK, 0 rows affected (0.12 sec)
```

```
mysql> DESC patient;
```

Field	Type	Null	Key	Default	Extra
Patient_ID	int	NO	PRI	NULL	auto_increment
Full_Name	varchar(100)	NO		NULL	
Date_of_Birth	date	NO		NULL	
Gender	varchar(10)	NO		NULL	
Phone_Number	varchar(15)	NO	UNI	NULL	
Address	varchar(100)	NO		NULL	

6 rows in set (0.06 sec)

```
mysql> CREATE TABLE doctor (
  -> Doctor_ID INT AUTO_INCREMENT PRIMARY KEY,
  -> Full_Name VARCHAR(100) NOT NULL,
  -> Specialization VARCHAR(50) NOT NULL,
  -> Phone_Number VARCHAR(15) NOT NULL UNIQUE,
  -> Treatment_Cost DECIMAL(10,2) NOT NULL,
  -> Department VARCHAR(50) NOT NULL
  -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> DESC doctor;
```

Field	Type	Null	Key	Default	Extra
Doctor_ID	int	NO	PRI	NULL	auto_increment
Full_Name	varchar(100)	NO		NULL	
Specialization	varchar(50)	NO		NULL	
Phone_Number	varchar(15)	NO	UNI	NULL	
Treatment_Cost	decimal(10,2)	NO		NULL	
Department	varchar(50)	NO		NULL	

6 rows in set (0.00 sec)

```
mysql> CREATE TABLE appointment (
  -> Appointment_ID INT AUTO_INCREMENT PRIMARY KEY,
  -> Patient_ID INT NOT NULL,
  -> Doctor_ID INT NOT NULL,
  -> Appointment_DateTime DATETIME NOT NULL,
  -> Status VARCHAR(20) NOT NULL,
  -> FOREIGN KEY (Patient_ID) REFERENCES patient(Patient_ID),
  -> FOREIGN KEY (Doctor_ID) REFERENCES doctor(Doctor_ID)
  -> );
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> DESC appointment;
```

Field	Type	Null	Key	Default	Extra
Appointment_ID	int	NO	PRI	NULL	auto_increment
Patient_ID	int	NO	MUL	NULL	
Doctor_ID	int	NO	MUL	NULL	
Appointment_DateTime	datetime	NO		NULL	
Status	varchar(20)	NO		NULL	

5 rows in set (0.00 sec)

```
mysql> INSERT INTO patient (Full_Name, Date_of_Birth, Gender, Phone_Number, Address)
  -> VALUES
  -> ('Rahul Sharma','2001-05-15','Male','9876543210','Pune'),
  -> ('Priya Patil','1994-08-22','Female','9876501234','Mumbai'),
  -> ('Amit Kumar','1979-02-10','Male','9876512345','Kolhapur'),
  -> ('Sneha Joshi','1996-11-05','Female','9876523456','Satara'),
  -> ('Rohan Desai','1987-07-18','Male','9876534567','Sangli'),
  -> ('Anjali Mulkarni','1990-03-25','Female','9876545678','Nashik'),
  -> ('Vikram Singh','1974-09-14','Male','9876556789','Nagpur'),
  -> ('Neha Verma','1991-12-30','Female','9876567890','Solapur'),
  -> ('Karan Mehta','1984-04-08','Male','9876578901','Aurangabad'),
  -> ('Pooja More','1993-10-12','Female','9876589012','Ahmednagar');
Query OK, 10 rows affected (0.03 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM patient;
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address
1	Rahul Sharma	2001-05-15	Male	9876543210	Pune
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli
6	Anjali Kulkarni	1998-03-25	Female	9876545678	Nashik
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur
8	Neha Verma	1991-12-30	Female	9876567890	Solapur
9	Karan Mehta	1984-04-08	Male	9876578901	Aurangabad
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar

```
10 rows in set (0.01 sec)

mysql> INSERT INTO doctor (Full_Name, Specialization, Phone_Number, Treatment_Cost, Department)
-> VALUES
-> ('Dr. Sharma','Cardiology','9876600001',5000.00,'Heart'),
-> ('Dr. Mehta','Orthopedic','9876600002',3500.00,'Bone'),
-> ('Dr. Patil','Neurology','9876600003',6500.00,'Brain'),
-> ('Dr. Joshi','Dermatology','9876600004',2500.00,'Skin'),
-> ('Dr. Kulkarni','Cardiology','9876600005',5500.00,'Heart'),
-> ('Dr. Deshmukh','Pediatrics','9876600006',3000.00,'Children'),
-> ('Dr. Shah','ENT','9876600007',2800.00,'ENT'),
-> ('Dr. Reddy','General Medicine','9876600008',2000.00,'Medicine');
Query OK, 8 rows affected (0.01 sec)
Records: 8 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM doctor;
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Department
1	Dr. Sharma	Cardiology	9876600001	5000.00	Heart
2	Dr. Mehta	Orthopedic	9876600002	3500.00	Bone
3	Dr. Patil	Neurology	9876600003	6500.00	Brain
4	Dr. Joshi	Dermatology	9876600004	2500.00	Skin
5	Dr. Kulkarni	Cardiology	9876600005	5500.00	Heart
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00	Children
7	Dr. Shah	ENT	9876600007	2800.00	ENT
8	Dr. Reddy	General Medicine	9876600008	2000.00	Medicine

```
8 rows in set (0.01 sec)

mysql> INSERT INTO appointment (Patient_ID, Doctor_ID, Appointment_DateTime, Status)
-> VALUES
-> (1,1,'2026-08-02 09:00:00','Scheduled'),
-> (2,2,'2026-08-02 10:00:00','Completed'),
-> (3,3,'2026-08-03 11:00:00','Scheduled'),
-> (4,4,'2026-08-03 12:00:00','Cancelled'),
-> (5,5,'2026-08-04 09:30:00','Scheduled'),
-> (6,6,'2026-08-04 10:30:00','Completed'),
-> (7,7,'2026-08-05 11:30:00','Scheduled'),
-> (8,8,'2026-08-05 12:30:00','Cancelled'),
-> (9,1,'2026-08-06 09:15:00','Scheduled'),
-> (10,2,'2026-08-06 10:15:00','Scheduled');
Query OK, 10 rows affected (0.02 sec)
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM appointment;
```

Appointment_ID	Patient_ID	Doctor_ID	Appointment_DateTime	Status
1	1	1	2026-08-02 09:00:00	Scheduled
2	2	2	2026-08-02 10:00:00	Completed
3	3	3	2026-08-03 11:00:00	Scheduled
4	4	4	2026-08-03 12:00:00	Cancelled
5	5	5	2026-08-04 09:30:00	Scheduled
6	6	6	2026-08-04 10:30:00	Completed
7	7	7	2026-08-05 11:30:00	Scheduled
8	8	8	2026-08-05 12:30:00	Cancelled
9	9	1	2026-08-06 09:15:00	Scheduled
10	10	2	2026-08-06 10:15:00	Scheduled

```
10 rows in set (0.00 sec)
```

Q1.Delete a particular patient record using P_ID.

```
mysql> DELETE FROM appointment
-> WHERE Patient_ID = 9;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> DELETE FROM patient
-> WHERE Patient_ID = 9;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT * FROM patient;
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address
1	Rahul Sharma	2001-05-15	Male	9876543210	Pune
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli
6	Anjali Kulkarni	1998-03-25	Female	9876545678	Nashik
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur
8	Neha Verma	1991-12-30	Female	9876567890	Solapur
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar

```
9 rows in set (0.00 sec)
```

Q2.Display all patients in descending order of age.

```
mysql> select * from patient order by age desc;
ERROR 1054 (42S22): Unknown column 'age' in 'order clause'
mysql> ALTER TABLE patient
-> ADD Age INT;
```

```
Query OK, 0 rows affected (0.03 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
mysql>
mysql> UPDATE patient
-> SET Age = TIMESTAMPDIFF(YEAR, Date_of_Birth, CURDATE());
Query OK, 9 rows affected (0.01 sec)
Rows matched: 9 Changed: 9 Warnings: 0
```

```
mysql> SELECT *
-> FROM patient
-> ORDER BY Age DESC;
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29
6	Anjali Kulkarni	1998-03-25	Female	9876545678	Nashik	28
1	Rahul Sharma	2001-05-15	Male	9876543210	Pune	25

```
9 rows in set (0.00 sec)
```

Q3.Display all patients in descending order of treatment cost.

```
mysql> SELECT *
-> FROM doctor
-> ORDER BY Treatment_Cost DESC;
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost	Department
3	Dr. Patil	Neurology	9876600003	6500.00	Brain
5	Dr. Kulkarni	Cardiology	9876600005	5500.00	Heart
1	Dr. Sharma	Cardiology	9876600001	5000.00	Heart
2	Dr. Mehta	Orthopedic	9876600002	3500.00	Bone
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00	Children
7	Dr. Shah	ENT	9876600007	2800.00	ENT
4	Dr. Joshi	Dermatology	9876600004	2500.00	Skin
8	Dr. Reddy	General Medicine	9876600008	2000.00	Medicine

```
8 rows in set (0.00 sec)
```

Q4.Change (update) a patient's contact number.

```
mysql> update patient set Phone_Number='999999999' where Patient_ID=1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from patient;
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39
6	Anjali Kulkarni	1998-03-25	Female	9876545678	Nashik	28
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32

```
9 rows in set (0.00 sec)
```

Q5.Drop (delete) any column from the Doctor table.

```
mysql> ALTER TABLE doctor
-> DROP COLUMN Doctor_ID;
ERROR 1829 (HY000): Cannot drop column 'Doctor_ID': needed in a foreign key constraint 'appointment_ibfk_2' of table 'appointment'
mysql> ALTER TABLE doctor
-> DROP COLUMN Department;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from doctor;
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost
1	Dr. Sharma	Cardiology	9876600001	5000.00
2	Dr. Mehta	Orthopedic	9876600002	3500.00
3	Dr. Patil	Neurology	9876600003	6500.00
4	Dr. Joshi	Dermatology	9876600004	2500.00
5	Dr. Kulkarni	Cardiology	9876600005	5500.00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00
7	Dr. Shah	ENT	9876600007	2800.00
8	Dr. Reddy	General Medicine	9876600008	2000.00

```
8 rows in set (0.00 sec)
```

Q6.Update a patient's address and contact number.

```
mysql> update patient set Address='Nagpur',Phone_Number='8888888888' where Patient_ID=6;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from patient
```

```
-> ;
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39
6	Anjali Kulkarni	1998-03-25	Female	8888888888	Nagpur	28
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32

```
9 rows in set (0.00 sec)
```

Q7.Increase the treatment cost by ₹100 for a particular treatment.

```
mysql> UPDATE doctor
-> SET Treatment_Cost = Treatment_Cost + 100
-> WHERE Specialization='ENT';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> select * from doctor;
```

Doctor_ID	Full_Name	Specialization	Phone_Number	Treatment_Cost
1	Dr. Sharma	Cardiology	9876600001	5000.00
2	Dr. Mehta	Orthopedic	9876600002	3500.00
3	Dr. Patil	Neurology	9876600003	6500.00
4	Dr. Joshi	Dermatology	9876600004	2500.00
5	Dr. Kulkarni	Cardiology	9876600005	5500.00
6	Dr. Deshmukh	Pediatrics	9876600006	3000.00
7	Dr. Shah	ENT	9876600007	2900.00
8	Dr. Reddy	General Medicine	9876600008	2000.00

```
8 rows in set (0.00 sec)
```

Q8.Alter the Patient table by adding a new Date column.

```
mysql> ALTER TABLE patient
-> ADD Admission_Date DATE;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-01' WHERE Patient_ID=1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-03' WHERE Patient_ID=2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-05' WHERE Patient_ID=3;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-08' WHERE Patient_ID=4;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-10' WHERE Patient_ID=5;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-12' WHERE Patient_ID=6;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-15' WHERE Patient_ID=7;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-18' WHERE Patient_ID=8;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE patient SET Admission_Date='2026-07-20' WHERE Patient_ID=9;
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0
```

```
mysql> UPDATE patient SET Admission_Date='2026-07-22' WHERE Patient_ID=10;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM patient;
+-----+-----+-----+-----+-----+-----+-----+-----+
| Patient_ID | Full_Name | Date_of_Birth | Gender | Phone_Number | Address | Age | Admission_Date |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | Rahul Sharma | 2001-05-15 | Male | 9999999999 | Pune | 25 | 2026-07-01 |
| 2 | Priya Patil | 1990-08-22 | Female | 9876501234 | Mumbai | 31 | 2026-07-03 |
| 3 | Amit Kumar | 1979-02-10 | Male | 9876512345 | Kolhapur | 47 | 2026-07-05 |
| 4 | Sneha Joshi | 1996-11-05 | Female | 9876523456 | Satara | 29 | 2026-07-08 |
| 5 | Rohan Desai | 1987-07-18 | Male | 9876534567 | Sangli | 39 | 2026-07-10 |
| 6 | Anjali Kulkarni | 1998-03-25 | Female | 8888888888 | Nagpur | 28 | 2026-07-12 |
| 7 | Vikram Singh | 1974-09-14 | Male | 9876556789 | Nagpur | 51 | 2026-07-15 |
| 8 | Neha Verma | 1991-12-30 | Female | 9876567890 | Solapur | 34 | 2026-07-18 |
| 10 | Pooja More | 1993-10-12 | Female | 9876589012 | Ahmednagar | 32 | 2026-07-22 |
+-----+-----+-----+-----+-----+-----+-----+-----+
9 rows in set (0.00 sec)
```

Q9.Alter the Doctor table by adding a new Time column.

```
mysql> alter table doctor add Visiting_Time TIME;
Query OK, 0 rows affected (0.02 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='09:00:00' WHERE Doctor_ID=1;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='09:30:00' WHERE Doctor_ID=2;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='10:00:00' WHERE Doctor_ID=3;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='10:30:00' WHERE Doctor_ID=4;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='11:00:00' WHERE Doctor_ID=5;
```

```
mysql> UPDATE doctor SET Visiting_Time='11:00:00' WHERE Doctor_ID=5;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='11:30:00' WHERE Doctor_ID=6;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='12:00:00' WHERE Doctor_ID=7;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> UPDATE doctor SET Visiting_Time='12:30:00' WHERE Doctor_ID=8;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0

mysql> select * from doctor;
+-----+-----+-----+-----+-----+-----+
| Doctor_ID | Full_Name | Specialization | Phone_Number | Treatment_Cost | Visiting_Time |
+-----+-----+-----+-----+-----+-----+
| 1 | Dr. Sharma | Cardiology | 9876600001 | 5000.00 | 09:00:00 |
| 2 | Dr. Mehta | Orthopedic | 9876600002 | 3500.00 | 09:30:00 |
| 3 | Dr. Patil | Neurology | 9876600003 | 6500.00 | 10:00:00 |
| 4 | Dr. Joshi | Dermatology | 9876600004 | 2500.00 | 10:30:00 |
| 5 | Dr. Kulkarni | Cardiology | 9876600005 | 5500.00 | 11:00:00 |
| 6 | Dr. Deshmukh | Pediatrics | 9876600006 | 3000.00 | 11:30:00 |
| 7 | Dr. Shah | ENT | 9876600007 | 2900.00 | 12:00:00 |
| 8 | Dr. Reddy | General Medicine | 9876600008 | 2000.00 | 12:30:00 |
+-----+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)
```

Q10.Modify the data type of the Phone No. column in any table.

```
mysql> ALTER TABLE patient
-> MODIFY Phone_Number VARCHAR(20);
Query OK, 0 rows affected (0.07 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> select * from patient;
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25	2026-07-01
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31	2026-07-03
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47	2026-07-05
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29	2026-07-08
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39	2026-07-10
6	Anjali Kulkarni	1998-03-25	Female	8888888888	Nagpur	28	2026-07-12
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51	2026-07-15
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34	2026-07-18
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32	2026-07-22

```
9 rows in set (0.00 sec)
```

Q11.Display all patients who are older than 30 years.

```
mysql>
mysql> select from patient where age>30;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'f
rom patient where age>30' at line 1
mysql> select * from patient where age>30;
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31	2026-07-03
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47	2026-07-05
5	Rohan Desai	1987-07-18	Male	9876523456	Sangli	39	2026-07-10
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51	2026-07-15
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34	2026-07-18
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32	2026-07-22

```
6 rows in set (0.00 sec)
```

Q12.Retrieve doctor names whose treatment cost is greater than ₹2,000.

```
mysql> select Full_Name from doctor where Treatment_Cost >2000;
```

Full_Name
Dr. Sharma
Dr. Mehta
Dr. Patil
Dr. Joshi
Dr. Kulkarni
Dr. Deshmukh
Dr. Shah

```
7 rows in set (0.01 sec)
```

Q13.Count the total number of patients in the hospital.

```
mysql> SELECT COUNT(*) AS Total_Patients
-> FROM patient;
```

Total_Patients
9

```
1 row in set (0.01 sec)
```

Q14.Find the maximum treatment cost among all doctors.

```
mysql> SELECT MAX(Treatment_Cost) AS Maximum_Treatment_Cost
-> FROM doctor;
```

Maximum_Treatment_Cost
6500.00

```
1 row in set (0.01 sec)
```

Q15.Display all patients sorted by age in descending order.

```
mysql> select * from patient order by age desc;
```

Patient_ID	Full_Name	Date_of_Birth	Gender	Phone_Number	Address	Age	Admission_Date
7	Vikram Singh	1974-09-14	Male	9876556789	Nagpur	51	2026-07-15
3	Amit Kumar	1979-02-10	Male	9876512345	Kolhapur	47	2026-07-05
5	Rohan Desai	1987-07-18	Male	9876534567	Sangli	39	2026-07-10
8	Neha Verma	1991-12-30	Female	9876567890	Solapur	34	2026-07-18
10	Pooja More	1993-10-12	Female	9876589012	Ahmednagar	32	2026-07-22
2	Priya Patil	1994-08-22	Female	9876501234	Mumbai	31	2026-07-03
4	Sneha Joshi	1996-11-05	Female	9876523456	Satara	29	2026-07-08
6	Anjali Wulkarni	1998-03-25	Female	8888888888	Nagpur	28	2026-07-12
1	Rahul Sharma	2001-05-15	Male	9999999999	Pune	25	2026-07-01

```
9 rows in set (0.00 sec)
```

```
mysql> UPDATE appointment
-> SET Status='Completed'
-> WHERE Appointment_ID=1;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
mysql> SELECT * FROM appointment;
```

Appointment_ID	Patient_ID	Doctor_ID	Appointment_DateTime	Status
1	1	1	2026-08-02 09:00:00	Completed
2	2	2	2026-08-02 10:00:00	Completed
3	3	3	2026-08-03 11:00:00	Scheduled
4	4	4	2026-08-03 12:00:00	Cancelled
5	5	5	2026-08-04 09:30:00	Scheduled
6	6	6	2026-08-04 10:30:00	Completed
7	7	7	2026-08-05 11:30:00	Scheduled
8	8	8	2026-08-05 12:30:00	Cancelled
10	10	2	2026-08-06 10:15:00	Scheduled

```
9 rows in set (0.00 sec)
```

Learning Outcome:

After completing this assignment, the following concepts I learned:

- Creating databases and tables using SQL.
- Applying SQL constraints such as Primary Key, Foreign Key, Unique, Not Null, and Auto Increment.
- Inserting, updating, deleting, and retrieving records.
- Modifying table structures using ALTER and MODIFY commands.
- Sorting records using ORDER BY.
- Using aggregate functions like MAX() and COUNT().
- Maintaining relationships between multiple tables using Foreign Keys.
- Designing a simple Hospital Management System database.

Conclusion:

The Hospital Management System was successfully implemented using SQL. The database efficiently stores patient, doctor, and appointment information while maintaining data integrity through various constraints. Different SQL operations such as CREATE, INSERT, UPDATE, DELETE, ALTER, and SELECT were performed successfully. This assignment helped in understanding relational database design, table relationships, constraints, and basic SQL query execution, providing practical knowledge of Database Management Systems (DBMS).